

Bayesian inference using Catalytic prior distributions for Cox regression models

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Joint work with Dongming Huang

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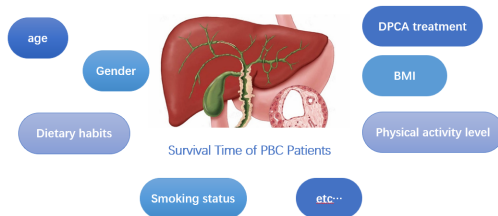
Dec 6, 2023

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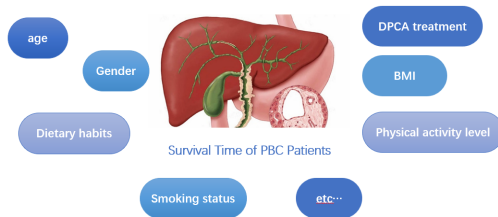
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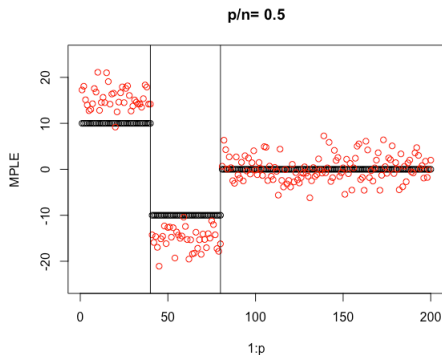
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coxph(Surv(time, status) ~ X)
```

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- Cox regression model: popular way in survival analysis

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- [Zhang et al., 2022] $p = 200, n = 400$



Introduction of Cox regression model

- Observed data: $\mathbf{D} = \{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n$, $\mathbf{X}_i \in \mathbb{R}^p$ is the feature vector, δ_i is the censoring indicator, and $Y_i = \min(T_i, C_i)$, where T_i is the survival time for an uncensored subject and C_i is the censoring time for a censored subject.

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- Cox regression model: semi-parametric

$$L(\boldsymbol{\beta}, h_0 \mid \mathbf{D}) = \prod_{i=1}^n \left\{ \exp(\mathbf{X}_i' \boldsymbol{\beta}) h_0(Y_i) \right\}^{\delta_i} \exp \left\{ - \exp(\mathbf{X}_i' \boldsymbol{\beta}) H_0(Y_i) \right\}$$

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- $h_0(t)$: baseline hazard function, unknown nuisance parameter
- Partial likelihood [Cox, 1975]: MPLE maximizes

$$PL(\beta) = \prod_{i=1}^n \left\{ \frac{\exp(\mathbf{X}_i' \beta)}{\sum_{l \in \mathbf{R}_i} \exp(\mathbf{X}_l' \beta)} \right\}^{\delta_i}$$

where $\mathbf{R}_i := \{j : Y_j \geq Y_i\}$ is the collection of index for those are risky at time Y_i .

Recap the problem of classical method

- When dimension is large or insufficient sample, MPLE suffers
 - ① highly variable
 - ② not exist
 - ③ large bias

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- When dimension is large or insufficient sample, MPLE suffers
 - 1 highly variable
 - 2 not exist
 - 3 large bias
- Similar problems for MLE

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \left(\prod_i^n f(Y_i | \mathbf{x}_i, \theta) \right)$$

- large variability for large p
- MLE may not exist, e.g.: Logistic regression
- large bias

Related methods to address issues

- Existing strategy for MLE:
 - [Huang et al., 2020] proposed class of automatic priors called **catalytic prior** for GLM to provide stable estimation in high dimensional model.

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- Catalytic prior based on synthetic data

$$\pi_{cat,M}(\boldsymbol{\theta} \mid \tau) \propto \left(\prod_{i=1}^M f(Y_i^* \mid \mathbf{X}_i^*, \boldsymbol{\theta}) \right)^{\tau/M}$$

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- Posterior distribution:

$$\begin{aligned} \pi(\boldsymbol{\theta} \mid \{(Y_i, \mathbf{X}_i)\}_{i=1}^n) &\propto \left(\prod_i^n f(Y_i \mid \mathbf{X}_i, \boldsymbol{\theta}) \right) \pi_{cat,M}(\boldsymbol{\theta} \mid \tau) \\ &\propto \exp \left\{ \sum_{i=1}^n \log(f(Y_i \mid \mathbf{X}_i, \boldsymbol{\theta})) + \frac{\tau}{M} \sum_{i=1}^M \log(f(Y_i^* \mid \mathbf{X}_i^*, \boldsymbol{\theta})) \right\} \end{aligned}$$

Stable estimation for simpler model

- Experts focus on several important covariates.
E.g.: thyroid-related diseases—gender, emotional state, age, etc.

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Only fit simpler model with subset of covariates $\{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3\}$ ¹.

$\rightarrow \hat{\psi} = (\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, 0, \dots, 0) \rightarrow$ predictive distribution $g_*(y | \mathbf{x})$

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- Create fake data to "increase" sample size. M synthetic data points $\{(Y_i^*, \mathbf{X}_i^*)\}_{i=1}^M$ can be generated according to following strategy:

$$\mathbf{X}_i^* \stackrel{i.i.d.}{\sim} Q(\mathbf{x}), \quad Y_i^* | \mathbf{X}_i^* \sim g_*(y | \mathbf{X}_i^*)$$

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- Now we are able to define catalytic prior for β

$$\pi_{cat,M}(\boldsymbol{\theta} | \tau) \propto \left(\prod_{i=1}^M f(Y_i^* | \mathbf{X}_i^*, \boldsymbol{\theta}) \right)^{\tau/M}$$

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Generation of synthetic survival data

- Synthetic covariates: $\mathbf{X}_i^* \stackrel{i.i.d.}{\sim} Q(\mathbf{x})$
- Synthetic survival time:
 - 1 Simpler predictive model $g(y \mid \mathbf{X}, \delta, \psi)$ with ψ can be stably fitted from the observed data $\{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n$, denote the predictive distribution as $g_*(y \mid \mathbf{x}, \{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n)$
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Example:

- 1 Simpler model:

$$L(\psi | \{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n) = \prod_{i=1}^n \psi^{\delta_i} \exp\{-\psi Y_i\}, \quad \psi > 0.$$

- 2 $Y_i^* | \mathbf{X}_i^* \sim \text{Exp}(\hat{\psi})$

Natural extension: Catalytic prior on (β, h_0)

$$\mathbf{D} = \{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n, \mathbf{D}^* = \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M$$

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- Efficient point estimation for β : profile $h_0(\cdot)$ out
[Murphy and Van der Vaart, 2000]

Weighted mixture estimator

Algorithm: Compute **W**eighted **M**ixture estimator in R

Require: R package survival

Input: $D = \{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n$, $D^* = \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M$, $\tau > 0$

- ① Combine data:
 $\tilde{\mathbb{X}} \leftarrow \text{rbind}(\mathbb{X}, \mathbb{X}^*)$
 $\tilde{\mathbf{Y}} \leftarrow c(\mathbf{Y}, \mathbf{Y}^*)$
 $\tilde{\boldsymbol{\delta}} \leftarrow c(\boldsymbol{\delta}, \mathbf{1})$
- ② Compute weight vector:
 $\tilde{\mathbf{w}} \leftarrow c(\text{rep}(1, n), \text{rep}(\frac{\tau}{M}, M))$
- ③ Fit proportional hazards regression model with combined data and weight vector:
 $\text{fit} \leftarrow \text{coxph}(\text{Surv}(\tilde{\mathbf{Y}}, \tilde{\boldsymbol{\delta}}) \sim \tilde{\mathbb{X}}, \text{weights} = \tilde{\mathbf{w}})$
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Property:

- $\hat{\beta}_{WM, \tau}$ is consistent when p is fixed and $\tau = o(n)$.

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$$\begin{aligned}\pi_{\text{cox,cat}}(\beta \mid \tau) &\propto L\left(\beta, h_0^+ \mid \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M\right)^{\tau/M} \\ &= \left[\prod_{i=1}^M \left\{ \exp(\mathbf{X}_i^{*'} \beta) h_0^+ \right\} \exp \left\{ - \exp(\mathbf{X}_i^{*'} \beta) Y_i^* h_0^+ \right\} \right]^{\tau/M}\end{aligned}$$

- User-specific surrogate baseline hazard constant $h_0^+ > 0$
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 - 2 Computed in a data driven way.

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- User-specific surrogate baseline hazard constant $h_0^+ > 0$
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h_0^+ acts merely as a surrogate for the nuisance component to facilitate the construction of our catalytic prior. It does not need to be correctly specified or unbiased.

- Example on h_0^+ :

① $h_0^+ = \hat{\psi}$

$$\hat{\psi} = \arg \max_{\psi} \left(\prod_{i=1}^n \psi^{\delta_i} \exp \{ -\psi Y_i \} \right), \quad \psi > 0.$$

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- Catalytic prior for β :

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Bayesian Cox model with catalytic prior

$$\pi_{\text{cox,cat}}(\boldsymbol{\beta} \mid \tau) = L\left(\boldsymbol{\beta}, h_0^+ \mid \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M\right)^{\tau/M} \quad (1)$$

Procedure:

- Standard Bayesian Cox model with $L(\boldsymbol{\beta}, h_0)$ as likelihood and default prior for parameter $\pi_0(\boldsymbol{\beta}) \cdot \pi(h_0)$

$$\pi_{\text{post}}(\boldsymbol{\beta}, h_0 \mid \mathbf{D}) \propto L(\boldsymbol{\beta}, h_0) \cdot \pi_0(\boldsymbol{\beta}) \cdot \pi(h_0)$$

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- Replace default prior $\pi_0(\boldsymbol{\beta})$ on $\boldsymbol{\beta}$ by catalytic prior Eq.(1).
- Run MCMC to collect posterior samples.

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- Replace default prior $\pi_0(\beta)$ on β by catalytic prior Eq.(1).
- Run MCMC to collect posterior samples.

Property:

- $\pi_{\text{cox,cat}}(\beta \mid \tau)$ is proper under mild assumption.

Approximate posterior mode

$$\pi_{post}(\beta, h_0 \mid \mathbf{D}) = L(\beta, h_0) \cdot \pi_{cox,cat}(\beta \mid \tau) \cdot \pi(h_0)$$

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Logic: from [Sinha, 2003], when $\pi(h_0)$ is diffuse,

$$\pi_{margin}(\beta \mid \mathbf{D}) \approx PL(\beta) \cdot \pi_{cox,cat}(\beta \mid \tau)$$

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Point estimation (**C**atalytic **R**egularized estimator)

$$\hat{\beta}_{CR,\tau} = \arg \max_{\beta} \{ \log PL(\beta) + \log \pi_{cox,cat}(\beta \mid \tau) \}, \quad (2)$$

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- $\hat{\beta}_{CR,\tau}$ is consistent when p is fixed and $\tau = o(n)$.

Short summary

Method:

- Full Bayesian Procedure

$$\pi_{\text{cox,cat}}(\boldsymbol{\beta} \mid \tau) = L\left(\boldsymbol{\beta}, h_0^+ \mid \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M\right)^{\tau/M}$$

- Catalytic Regularized estimator

$$\hat{\boldsymbol{\beta}}_{CR,\tau} = \arg \max_{\boldsymbol{\beta}} \{\log PL(\boldsymbol{\beta}) + \log \pi_{\text{cox,cat}}(\boldsymbol{\beta} \mid \tau)\}$$

- Weighted Mixture estimator

$$\hat{\boldsymbol{\beta}}_{WM,\tau}$$

Theory:

- $\pi_{\text{cox,cat}}(\boldsymbol{\beta} \mid \tau)$ is proper
- $\hat{\boldsymbol{\beta}}_{CR,\tau}$ and $\hat{\boldsymbol{\beta}}_{WM,\tau}$ are consistent.

Numerical studies

Observed data:

- Covariates:

$$\mathbf{X}_{i,j} \sim \begin{cases} \text{Bernoulli}(0.1), & j = 1 \\ \chi_1^2, & j = 2 \\ \chi_4^2, & j = 3 \\ N(0, 1), & 4 \leq j \leq p \end{cases}$$

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- $n = 100$, 20% censored subjects.
- The true regression coefficient vector is set to be $\beta_0 = (4, -4, 3, -3, \mathbf{1}_{p-4})/\sqrt{p}$.
- T_i independently from an exponential distribution with a rate parameter of $0.5 \exp(\mathbf{X}_i^\top \beta_0)$.

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- $M=1000$
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Modification:

- half of the sampled X_1^* will be replaced by i.i.d. random variables draw from $\text{Bernoulli}(p = 0.5)$ —*flattening*
- continuous covariate ($j \geq 2$), half of sampled X_j^* will be replaced by i.i.d random variables draw from a normal distribution with median and interquartile range matching to those of the observed covariates.

Synthetic data:

- $M=1000$
- Synthetic covariates: independent resampling from original covariates

Modification:

- half of the sampled X_1^* will be replaced by i.i.d. random variables draw from $\text{Bernoulli}(p = 0.5)$ —*flattening*
- continuous covariate ($j \geq 2$), half of sampled X_j^* will be replaced by i.i.d random variables draw from a normal distribution with median and interquartile range matching to those of the observed covariates.
- Synthetic survival time: $Y_i^* \sim \text{Exp}(\hat{\psi})$

$$L(\psi \mid \{(\mathbf{X}_i, Y_i, \delta_i)\}_{i=1}^n) = \prod_{i=1}^n \psi^{\delta_i} \exp \{-\psi Y_i\}, \quad \psi > 0.$$

Numerical studies

Our methods: $h_0^+ = \hat{\psi}$

$$\pi_{\text{cox,cat}}(\boldsymbol{\beta} \mid \tau) = L\left(\boldsymbol{\beta}, h_0^+ \mid \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M\right)^{\tau/M}$$

- ① CRE: catalytic regularized estimator $\hat{\boldsymbol{\beta}}_{CR,\tau}$
- ② WME: weighted mixture estimator $\hat{\boldsymbol{\beta}}_{WM,\tau}$
- ③ CPM: posterior mean of MCMC sampler based on catalytic prior

Alternative methods:

$$\hat{\boldsymbol{\beta}}_\lambda = \arg \max_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \sum_{i=1}^n \delta_i \left(\mathbf{x}_i^\top \boldsymbol{\beta} - \log \sum_{j \in \mathcal{R}_i} \theta_j \right) - \lambda f(\boldsymbol{\beta}) \right\}$$

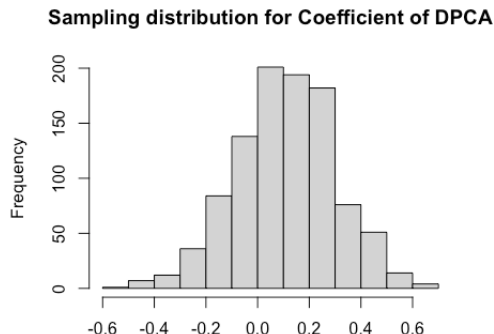
- ① $f(\boldsymbol{\beta}) = 0 \Rightarrow$ MPLE: maximum partial likelihood estimator
- ② $f(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_1 \Rightarrow$ Lasso
- ③ $f(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_2^2 \Rightarrow$ Ridge

$$\text{Deviance}(\beta_0, \hat{\beta}) = \left\{ \ell^{(\text{test})}(\beta_0) - \ell^{(\text{test})}(\hat{\beta}) \right\}$$

p	Methods	$\ \hat{\beta} - \beta_0\ ^2$	Predictive deviance
20	MPLE	0.95(0.06)	19.73(1.32)
	CRE (CV)	0.63(0.04)	12.87(0.80)
	CRE ($\tau = p$)	0.86(0.02)	19.53(0.70)
	WME (CV)	0.51 (0.02)	12.44 (0.75)
	CPM (CV)	0.79(0.04)	18.82(0.67)
	Ridge (CV)	0.58(0.03)	13.07(0.63)
	Lasso (CV)	0.75(0.04)	13.42(0.63)
40	MPLE	4.25(0.21)	103.38(4.99)
	CRE (CV)	0.82(0.02)	24.09(0.84)
	CRE ($\tau = p$)	0.88(0.01)	25.63(0.77)
	WME (CV)	0.76 (0.02)	23.37 (0.82)
	CPM (CV)	0.84(0.02)	24.63(0.76)
	Ridge (CV)	0.94(0.03)	24.53(0.87)
	Lasso (CV)	1.19(0.03)	24.06(0.92)

Real data: PBC dataset

- Data: $p = 18$, DPCA treatment + 17 covariates
- Full Bayesian analysis with $\tau = p$



95% Credible interval: $[-0.28, 0.47]$. Ineffectiveness of DPCA treatment has been reported [Locke III et al., 1996].

Summary

- Propose Cox catalytic prior on β and its corresponding approximate posterior mode.
- Derive weighted mixture estimator Using synthetic data to derive.
- Establish properness of prior and consistency of point estimation.
- Show proposed methods outperform classical MPLE and comparable with existing regularization method.

Summary

- Propose Cox catalytic prior on β and its corresponding approximate posterior mode.
- Derive weighted mixture estimator Using synthetic data to derive.
- Establish properness of prior and consistency of point estimation.
- Show proposed methods outperform classical MPLE and comparable with existing regularization method.

Future direction:

- Other statistical inference that involve partial likelihood or complex nuisance parameters.
- Other semi-parametric model.

<http://arxiv.org/abs/2312.01411>

Thank you!

Appendix: Prior on τ

If a fully Bayesian perspective on τ is adopted, we can impose a joint prior on (τ, β) as follows. Given any two positive scalar hyperparameters α and γ , we define a joint catalytic prior for (τ, β) as

$$\pi_{\alpha, \gamma}(\tau, \beta) \propto \Gamma_{\alpha, \gamma}(\tau) \cdot L\left(\beta, h_0^+ \mid \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M\right)^{\tau/M} \quad (3)$$

where $\Gamma_{\alpha, \gamma}(\tau)$ is a function defined as

$$\Gamma_{\alpha, \gamma}(\tau) = \tau^{p+\alpha-1} e^{-\tau(\kappa+\gamma^{-1})} \quad (4)$$

and

$$\kappa := \sup_{\beta \in \mathbb{R}^p} \frac{1}{M} \log L\left(\beta, h_0^+ \mid \{(\mathbf{X}_i^*, Y_i^*)\}_{i=1}^M\right) \quad (5)$$

Appendix: Ridge and Lasso

The ridge estimator is defined as follows:

$$\hat{\beta}_{\lambda} = \arg \max_{\beta \in \mathbb{R}^p} \left\{ \sum_{i=1}^n \delta_i \left(\mathbf{x}_i^{\top} \beta - \log \sum_{j \in \mathcal{R}_i} \theta_j \right) - \lambda f(\beta) \right\} \quad (6)$$

where $\theta_j = \exp(\mathbf{x}_j^{\top} \beta)$ and the penalty term $f(\beta) = \|\beta\|^2$, assuming that each entry of the covariates $\mathbf{x}_i$'s has been standardized to have zero mean and unit variance. The Lasso estimator is defined in a similar way as in (6) but with $f(\beta) = \|\beta\|_1$. Both the ridge and the Lasso estimates can be computed using the `glmnet` package in **R** [Simon et al., 2011].

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